# Day 15: Software Testing Concepts

Detailed Notes, Real-World Examples & Execution Flow

## 1. Core Testing Concepts

### 1. Smoke Testing (Build Verification Test)

**Concept:** Performed immediately after receiving a fresh build from the development team to check basic stability and critical features. It checks if the application is stable enough for detailed testing.

**When to Perform:** At the very first stage when a new build is deployed, before starting detailed manual or automated testing.

**Real-World Example:** Opening a mobile recharge app to check if the login screen loads and user credentials work without crashing the app. If login fails, the build is immediately rejected (Testing Stops).

### 2. Sanity Testing (Quick Impact Check)

**Concept:** Performed after minor code changes or quick bug fixes to quickly verify that the application is reasonably stable and immediate impacted areas work properly.

**When to Perform:** After Smoke Testing passes or when developers send a minor bug fix/code update.

**Key Note:** It is considered a *Subset of Regression Testing* because it focuses narrowly on impacted areas instead of testing the entire application.

**Real-World Example:** After a small update in the payment section of a recharge app, quickly verifying only the payment gateway and recharge screen without testing unrelated modules like user profile edits.

### 3. Re-testing (Bug Verification)

**Concept:** Testing a specific bug after it has been marked as "Fixed" by the developer to ensure the issue is completely resolved.

**When to Perform:** Whenever a reported bug is fixed by the developer.

**Key Note:** Only that specific bug and its related test data are checked. Surrounding features are not tested here.

**Real-World Example:** In a recharge module, entering an "Idea" mobile number was failing (Bug). Once fixed by the developer, we re-test by entering different Idea numbers to verify they are now accepted.

### 4. Regression Testing (Side-Effect Check)

**Concept:** Ensuring that recent code additions, bug fixes, or new features have not broken any existing, previously working features (old features).

**When to Perform:** After Re-testing and Sanity Testing are completed, right before releasing the final build to production.

**Types:**
• *Full Regression:* Testing all existing features across the whole app (best done via Automation).
• *Module/Impacted Regression:* Manually testing related sub-functionalities within a specific module after a major change.

**Real-World Example:** A new UPI payment method is added to the recharge app. Regression testing ensures that older payment methods (Debit Card, Net Banking) and past transaction history features still work smoothly without issues.

## 2. Order of Execution (Testing Flow)

The standard sequence followed during test execution in a software project:

| Order | Testing Type | Primary Objective | Next Step / Outcome |
|---|---|---|---|
| **Step 1** | Smoke Testing | Check overall build stability and core features. | If Pass $\rightarrow$ Proceed to Sanity. If Fail $\rightarrow$ Reject Build. |
| **Step 2** | Sanity Testing | Quickly verify minor fixes and immediate impacted areas. | Ensures surrounding features aren't broken by recent changes. |
| **Step 3** | Re-testing | Verify that specific reported bugs are completely fixed. | Confirms bug resolution using multiple test data inputs. |
| **Step 4** | Regression Testing | Ensure new changes haven't broken any existing functionalities. | Final safety net before releasing build to Production. |